SYSTEM DESIGN PACKAGE

NetGuardAI

Design Phase

Architecture and behavior designs with selection rationale for an AI-assisted network intrusion detection system.

1

Component diagram

Defines system modules and how they connect.

2

Data Flow Diagram

Traces traffic, processes, data stores, and outputs.

3

Activity diagram

Shows validation and benign-versus-attack logic.

4

Sequence diagram

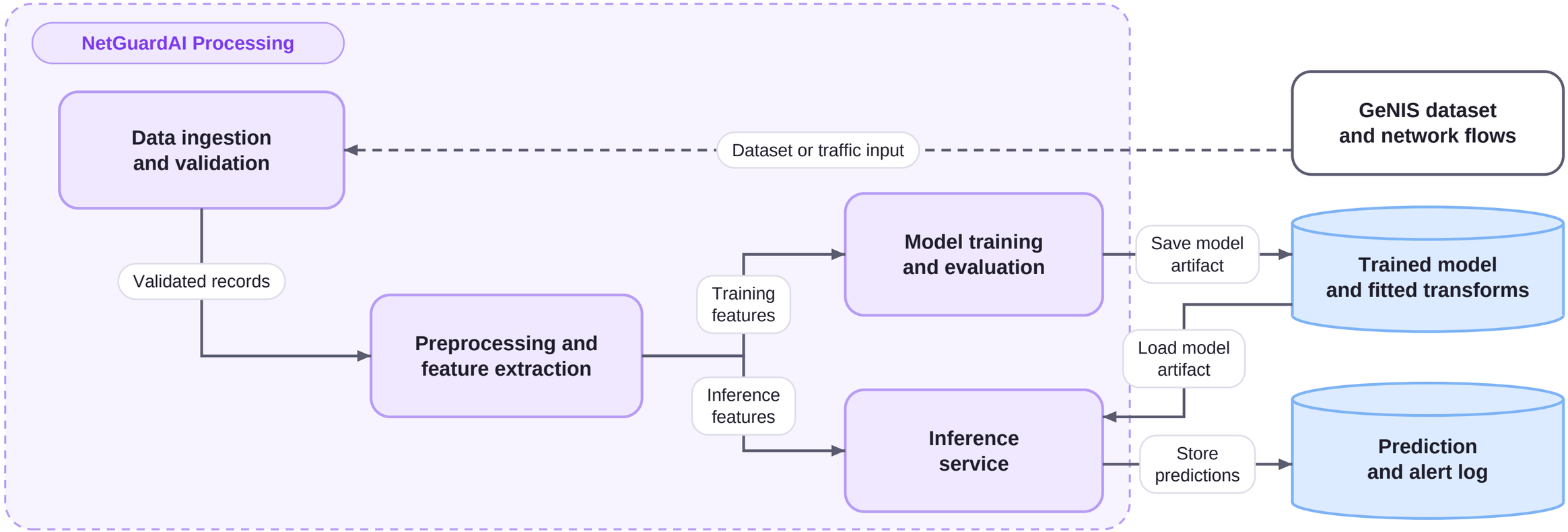
Shows message order for one traffic flow.

SCOPE NOTE

The Module 1 notebook covers offline GeNIS data ingestion, preprocessing, class reporting, and Random Forest baseline evaluation. Live traffic capture, the inference service, alerting, and the dashboard are proposed system-level components; include them in the final design only if they are in project scope.

1 Component Diagram

Structural design | Major components and dependencies



WHY THIS DESIGN WAS CHOSEN

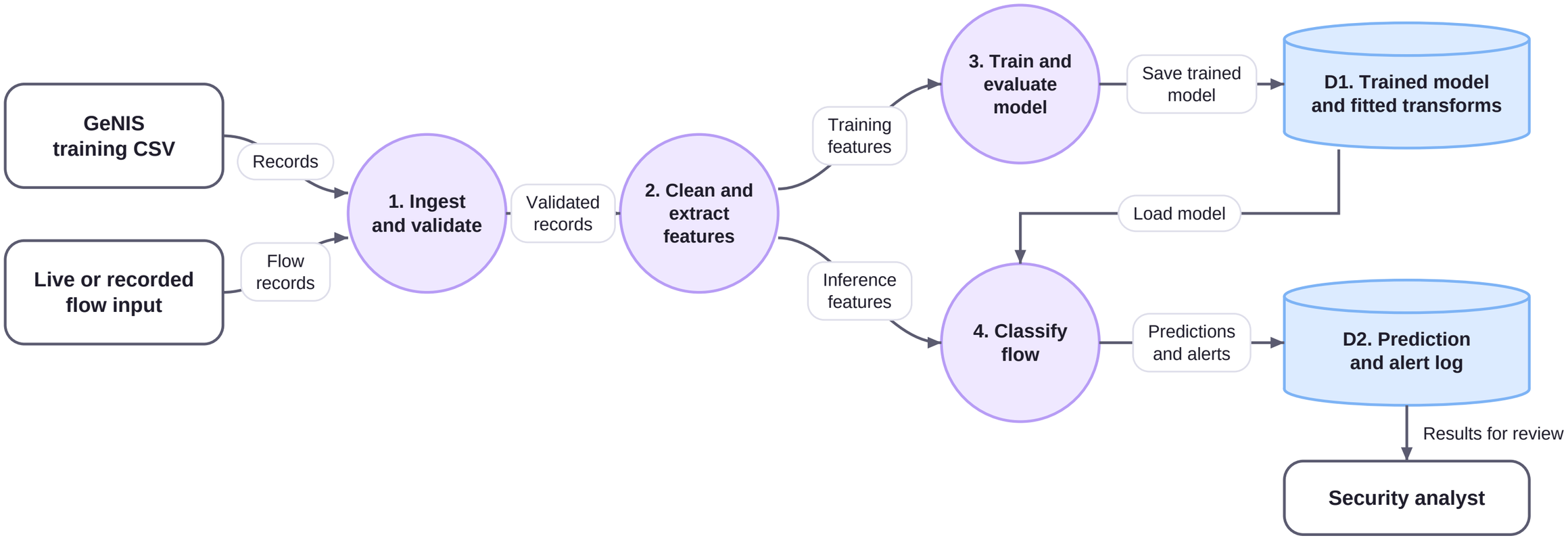
The component diagram gives a compact map of each system responsibility and the connections between them. It makes the offline training path distinct from the proposed inference path.

HOW IT FITS MODULE 1

Module 1 implements data loading, preprocessing, training, and evaluation. Live traffic inference and persistent alert logging are proposed extensions unless already implemented elsewhere.

2 Data Flow Diagram

Data boundaries | Inputs, processes, stores and outputs



WHY THIS DESIGN WAS CHOSEN

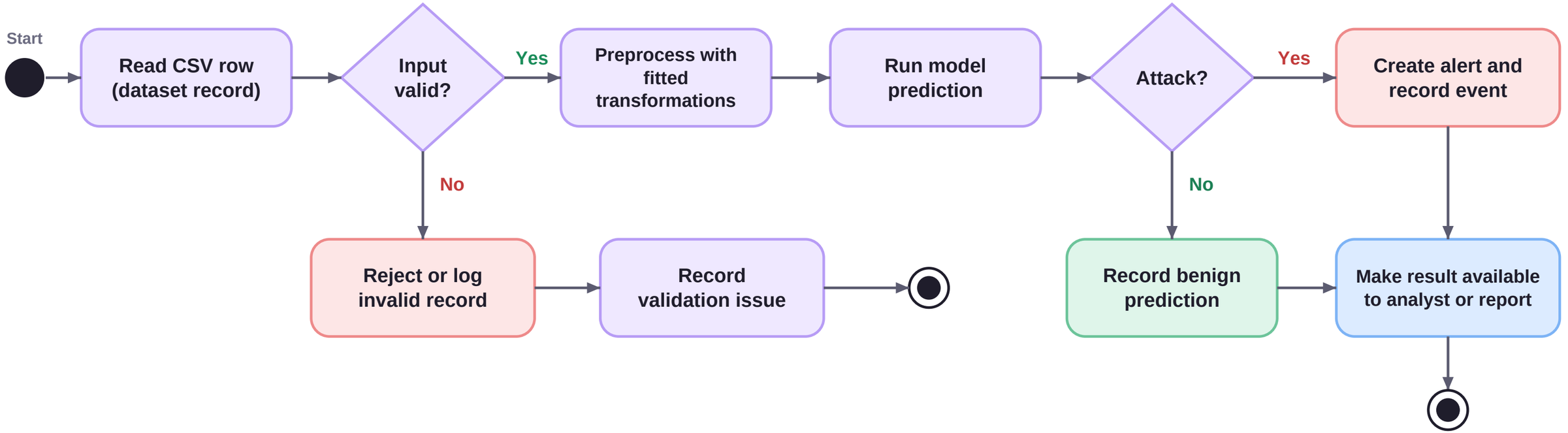
The DFD makes data boundaries visible: where traffic enters, which processes transform it, what data is stored, and who receives the results. It is useful for explaining the separation of training data, model artifacts, predictions, and alerts.

HOW IT FITS MODULE 1

The notebook covers GeNIS dataset ingestion, preprocessing, model training, and evaluation. Live-flow input and analyst-facing alert output are proposed deployment elements unless implemented elsewhere.

3 Activity Diagram

Behavioral design | Detection workflow and decision paths



WHY THIS DESIGN WAS CHOSEN

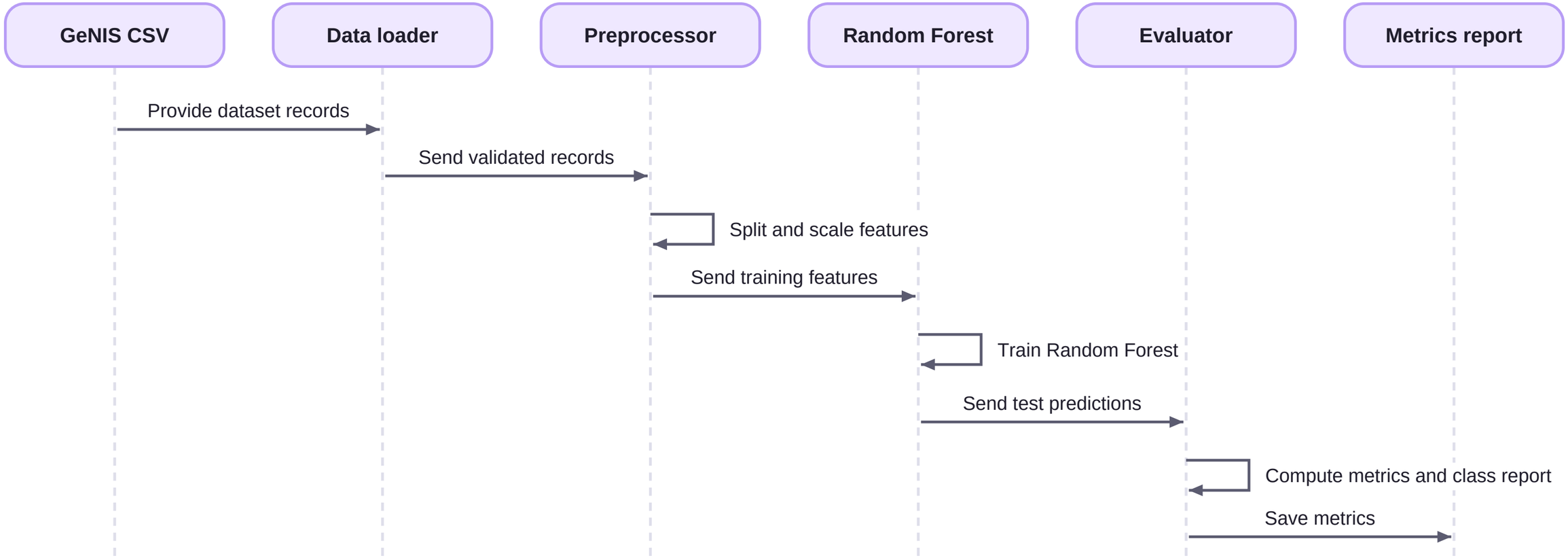
The activity diagram explains control flow. It shows the validation failure path and the two outcomes after classification, so a reader can follow what happens to both invalid and valid traffic.

HOW IT FITS MODULE 1

Module 1 processes a static GeNIS CSV offline, one record at a time through validation, preprocessing and classification. Alert creation and analyst presentation are proposed deployment behavior.

4 Sequence Diagram

Behavioral design | Offline training and evaluation sequence



WHY THIS DESIGN WAS CHOSEN

The sequence diagram clarifies who sends each message and in what order. It shows the offline path from loading the CSV, through preprocessing and training, to the saved evaluation results.

HOW IT FITS MODULE 1

This matches Module 1: the notebook loads the GeNIS CSV, splits and scales the features, trains a Random Forest baseline, evaluates it, and reports the metrics. Live inference is a future extension.